On Runs of Integers with Large Prime Factors: Sharp Bounds for $k(n)$

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Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \ldots, m+k$ divisible by some prime $> k$. We prove $k(n) \geq \exp((\log n)^{1/2-\varepsilon})$ for every fixed $\varepsilon > 0$ and all large n, unconditionally. We also prove $k(n) \leq \exp((\log n)^{1/2+\varepsilon})$ for every fixed $\varepsilon > 0$ and all large n, conditional on the short-interval smooth number asymptotic of Hildebrand–Tenenbaum [1] (Theorem 4 below, which we cite from the literature without reproving). Together these give $\log k(n) = (\log n)^{1/2+o(1)}$.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Let

$$\Psi(x, y) = \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. For positive integers k and n, a *k-run below n* is an interval $\{m+1, \ldots, m+k\}$ with $m \leq n$ such that $P^+(m+j) > k$ for every $1 \leq j \leq k$. Define $k(n)$ to be the maximum k for which a k-run below n exists.

2 Smooth Number Lemmas

Upper bound on Ψ: Rankin's method

Lemma 3 ([2, Theorem III.5.1]). *There is an effectively computable absolute constant $u_0 \geq 3$ such that for all $x \geq y \geq 2$ with $u = \log x / \log y \geq u_0$,*

$$\Psi(x, y) \ \leq \ x \cdot e^{-\frac{1}{2} u \log u}.$$

Proof. Cited from [2, Theorem III.5.1]. Rankin's method with the choice $\sigma = 1 - (\log u)/\log y$ gives $\Psi(x, y) \leq x\, e^{-u \log u + O(u/\log u)}$. The $O(u/\log u)$ term is at most $\frac{1}{2} u \log u$ once $\log^2 u \geq 2C$ for the implicit constant C; this holds for all $u \geq u_0$ with an effectively computable u_0. The full proof with all constants is in [2, Theorem III.5.1]. □

Short-interval asymptotics for smooth numbers

We state the deep result on which the upper bound rests. The following is the main result of Hildebrand–Tenenbaum [1] in the short-interval form presented in [2, Chapter III.5, Theorem 3].

Theorem 4 (Hildebrand–Tenenbaum [1, Théorème 1]; Tenenbaum [2, Ch. III.5, Theorem 3]). *Let ρ denote the Dickman function and let $\varepsilon > 0$ be fixed. Define*

$$\mathcal{R}_\varepsilon := \left\{(x, y) \in \mathbb{R}^2 : x \geq y \geq 2,\ \log y \geq (\log \log x)^{5/3+\varepsilon}\right\}. \tag{1}$$

For $(x, y) \in \mathcal{R}_\varepsilon$ write $u = \log x / \log y \geq 1$. Then

$$\Psi(x + y,\, y) - \Psi(x,\, y) \ = \ y\, \rho(u) \left(1 + \eta(x, y)\right), \tag{2}$$

where $\eta(x, y) \to 0$ uniformly for $(x, y) \in \mathcal{R}_\varepsilon$ as $y \to \infty$.

More precisely: for every $\delta > 0$ there exists $Y_0 = Y_0(\varepsilon, \delta)$, depending only on ε and δ (and not on x), such that

$$|\eta(x, y)| \ \leq \ \delta \qquad \text{for all } (x, y) \in \mathcal{R}_\varepsilon \text{ with } y \geq Y_0. \tag{3}$$

Remark 5. The uniformity in x expressed in (3)—that Y_0 is independent of x—is the key point for our application. This is exactly what is proved in [1, Théorème 1] and presented in [2, Ch. III.5, Theorem 3]: the asymptotic holds throughout the region $\mathcal{R}_\varepsilon$ simultaneously for all x, with an error controlled by y alone.

3 The Lower Bound

Theorem 6 (Lower bound)**.** *For every fixed $\varepsilon > 0$ and all sufficiently large n,*

$$k(n) \;\ge\; \exp\bigl((\log n)^{1/2-\varepsilon}\bigr).$$

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: The k_0-smooth integers in $[1, n]$ are sparse. Let $u_0 := \log n/\log k_0 = (\log n)^{1/2+\varepsilon}(1 + o(1)) \to \infty$. For all large n we have $u_0 \ge u_0^*$ (the constant from Lemma 3), so

$$\Psi(n, k_0) \;\le\; n \cdot e^{-\frac{1}{2}u_0 \log u_0}.$$

Now

$$\tfrac{1}{2}u_0 \log u_0 = \tfrac{1}{2}(\log n)^{1/2+\varepsilon}(1+o(1))\cdot(1/2+\varepsilon)\log\log n\,(1+o(1)) \;\ge\; (\log n)^{1/2+\varepsilon/2}$$

for all large n. Since $\log k_0 = (\log n)^{1/2-\varepsilon}$ and $(\log n)^{1/2+\varepsilon/2} \gg (\log n)^{1/2-\varepsilon}$, we obtain $\Psi(n, k_0) \le n/k_0$ for all sufficiently large n.

Step 2: Pigeonhole produces an empty block. Let $B := \lfloor n/k_0 \rfloor$. Partition $\{1, \dots, Bk_0\}$ into B consecutive blocks $I_i = [(i-1)k_0 + 1,\ ik_0]$. Since $\Psi(n, k_0) \le n/k_0 = B + O(1) < B$ for all large n, by pigeonhole some block I_i contains no k_0-smooth integer.

Step 3: The empty block is a k_0-run. Every $j \in I_i$ has $P^+(j) > k_0$. Set $m = (i-1)k_0 \le n$; then $\{m+1, \dots, m+k_0\}$ is a k_0-run below n, so $k(n) \ge k_0 \ge e^{(\log n)^{1/2-\varepsilon}(1+o(1))}$. □

4 The Upper Bound

Theorem 7 (Upper bound)**.** *Assume Theorem 4. For every fixed $\varepsilon > 0$ and all sufficiently large n,*

$$k(n) \;\le\; \exp\bigl((\log n)^{1/2+\varepsilon}\bigr).$$

Proof. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. We show every interval $[m+1, m+k_1]$ with $m \le n$ contains a k_1-smooth integer.

Case 2: $m < k_1$. Then $k_1 \in [m+1, m+k_1]$ and k_1 is k_1-smooth.

Case 1: $m \ge k_1$.

Step 1a: $(m,k_1)\in\mathcal{R}_\varepsilon$. (i) $m\ge k_1\ge 2$: holds by assumption. (ii) $\log k_1\ge(\log\log m)^{5/3+\varepsilon}$: since $m\le n$ we have $\log\log m\le\log\log n$, so it suffices that

$$(\log n)^{1/2+\varepsilon}(1+o(1)) \;\ge\; (\log\log n)^{5/3+\varepsilon}.$$

This holds for all large n since $(\log n)^{1/2+\varepsilon}/(\log\log n)^{5/3+\varepsilon}\to\infty$. Hence $(m,k_1)\in\mathcal{R}_\varepsilon$ for every $m\in[k_1,n]$ and all large n.

Step 1b: The main term is at least 2. Set $u_1:=\log m/\log k_1\in[1,(\log n)^{1/2-\varepsilon}(1+o(1))]$ for $m\in[k_1,n]$. Since ρ is decreasing, $k_1\rho(u_1)\ge k_1\rho((\log n)^{1/2-\varepsilon})$. The Dickman asymptotics $\rho(u)=e^{-u\log u(1+o(1))}$ [2, Corollary III.5.4] give

$$\log\bigl(k_1\rho((\log n)^{1/2-\varepsilon})\bigr)=(\log n)^{1/2+\varepsilon}(1+o(1))-(1/2-\varepsilon)(\log n)^{1/2-\varepsilon}\log\log n\,(1+o(1))\;\to\;+\infty,$$

since $(1/2-\varepsilon)(\log n)^{1/2-\varepsilon}\log\log n=o((\log n)^{1/2+\varepsilon})$. Hence there exists an effectively computable $n_1=n_1(\varepsilon)$ such that

$$k_1\,\rho(u_1)\;\ge\;2\quad\text{for all } m\in[k_1,n] \text{ and all } n\ge n_1. \tag{4}$$

Step 1c: Application of Theorem 4 with $\delta=\frac{1}{2}$. By Theorem 4 there exists $Y_0=Y_0(\varepsilon,\frac{1}{2})$, independent of m, such that $|\eta(m,k_1)|\le\frac{1}{2}$ for all $(m,k_1)\in\mathcal{R}_\varepsilon$ with $k_1\ge Y_0$. Since $k_1\to\infty$, there exists $n_0=n_0(\varepsilon)$ with $k_1\ge Y_0$ for all $n\ge n_0$. For $n\ge\max(n_0,n_1)$ and $m\in[k_1,n]$:

$$\Psi(m+k_1,k_1)-\Psi(m,k_1)=k_1\rho(u_1)(1+\eta(m,k_1))\;\ge\;\tfrac{1}{2}\,k_1\rho(u_1)\;\ge\;\tfrac{1}{2}\cdot 2=1.$$

So $[m+1,m+k_1]$ contains at least one k_1-smooth integer.

Conclusion. No k_1-run below n exists, so $k(n)<k_1\le e^{(\log n)^{1/2+\varepsilon}}$. □

Remark 8. The proof uses Theorem 4 in the following specific way: the error $\eta(m,k_1)$ must be $\le\frac{1}{2}$ simultaneously for all $m\in[k_1,n]$, with a threshold Y_0 independent of m. This is precisely the x-uniform form of (3). A bound valid only for fixed x would not suffice.

5 Main Result

Theorem 9. *For every fixed $\varepsilon>0$ and all sufficiently large n,*

$$\exp\bigl((\log n)^{1/2-\varepsilon}\bigr)\;\le\;k(n)\;\le\;\exp\bigl((\log n)^{1/2+\varepsilon}\bigr).$$

In particular, $\log k(n)=(\log n)^{1/2+o(1)}$. The lower bound is unconditional; the upper bound relies on Theorem 4, cited from [1, 2].

Proof. Theorems 6 and 7. □

References

[1] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.

[2] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 3rd ed., 2015.